THE NOVA RESEARCH CURRICULUM

Physics

Grade 7 · Term 2

Student edition

Outcomes: PH-7.2 waves and sound; PH-7.3 heat transfer and temperature scales.

Lesson notes

- **Waves (Weeks 1–4):** waves carry energy, not matter. Transverse (water, light) and longitudinal (sound). Amplitude, wavelength, frequency (hertz, Hz). Slinky demonstration.

  ▶ **Try it on NovaSchool:** Waves: amplitude, frequency and sound — open **Simulations** on your dashboard.

- **Sound (Weeks 5–8):** sound is made by vibrations and needs a medium; it cannot travel through a vacuum. Speed in air about 340 m/s; faster in water and solids. Higher frequency = higher pitch; bigger amplitude = louder. Echo; noise pollution and protecting hearing.
- **PH-7.3 Heat (Weeks 9–12):** conduction (solids, metal spoon), convection (liquids and gases, sea breeze), radiation (no medium needed, heat from the Sun). Conductors and insulators. Temperature scales: °C, °F, K; K = °C + 273; °F = (9/5 × °C) + 32. Water boils at 100 °C = 212 °F = 373 K.

Workbook pages

Page	Task
1	Label a wave: amplitude, wavelength.
2	String-telephone practical; pitch and loudness table.
3	Conduction, convection or radiation? Sort 12 examples.
4	Convert 10 temperatures.

Question bank

#	Outcome	Question	Your answer
1	PH-7.2	Sound cannot travel through …	water / steel / air / a vacuum
2	PH-7.2	Higher frequency gives a …	louder sound / higher pitch / slower sound / lower pitch
3	PH-7.2	Frequency is measured in …	metres / newtons / hertz / joules
4	PH-7.2	Sound is a longitudinal wave. True or false.	True / False
5	PH-7.3	Heat from the Sun reaches Earth by …	conduction / convection / radiation / evaporation
6	PH-7.3	A sea breeze is caused by …	conduction / convection / radiation / reflection

#	Outcome	Question	Your answer
7	PH-7.3	25 °C in kelvin?	25 K / 248 K / 298 K / 77 K
8	PH-7.3	100 °C in °F?	100 °F / 180 °F / 212 °F / 373 °F
9	PH-7.3	Why are cooking pots often given plastic handles?	_____